

Strategic Agent-Based Equilibrium Models for Urban Mobility: the Traffic Filter Location Problem

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Abstract. Urban planning is progressively shifting away from car-centric design and towards public transport due to a combination of environmental, societal, and economic factors. One method to encourage public transport usage is the installation of traffic filters – interventions that restrict certain modes of transport (such as cars) from parts of the network while allowing others (such as buses or trams) through. Traffic filters are being increasingly adopted by planning authorities worldwide, making it important to model them and formally study their impact. To capture the nuanced traffic and behavioural dynamics at play, we propose a multi-modal, stochastic user equilibrium model with elastic demands. We prove that an equilibrium always exists in this formulation and demonstrate how it can be found in practice by implementing a gap function-based column generation algorithm. We then use this model to study optimal traffic filter placement under a range of objective functions.

Keywords: Multimodal traffic assignment · Stochastic user equilibrium · Traffic simulation · Multiagent model design.

1 Introduction

Effective traffic flow modelling plays a crucial role in alleviating congestion, leading to a reduction in pollutant emissions stemming from fuel consumption, and fostering the sustainability of urban centres [43]. Historically, urban planning has been ‘car-centric’, with an emphasis on maximising the accessibility of urban environments to personal vehicles. However, in recent years there has been a growing shift away from car-centric design, placing an increasing emphasis on public transport links, walkability, and cycling infrastructure. The reduction of the number of private road vehicles, and the dependence on them, not only tackles economic and societal challenges, but also addresses environmental concerns.

From a mathematical standpoint, traffic flow models usually fall into one of two main categories: (i) a centralized controller acts as a single decision-making

entity that has the authority to dictate the actions of all agents, or (ii) each agent that travels through the network independently and selfishly selects its own actions. These two categories often lead to two distinct forms of equilibrium, known as Wardrop Equilibria [44]. The former category (i) may see the emergence of a System Optimal (SO), where the central planner allocates journeys to minimise the average travel time. The latter (ii) will likely result in a User Equilibrium (UE), where every agent travels by the shortest path available to it, but the overall average travel time is increased compared to the former case [34,11].

When it comes to urban areas, both paradigms have flaws. On the one hand, assuming that centralised authority completely controls and determines the traffic flow is often not implementable in practice. On the other hand, selfish agent based models may reflect observed driver behaviour more, but, in reality, agents choose between different modes of transport, or may decide not to travel at all. Additionally, travelling time is only one factor in planning a journey, among others such as cost of fuel or bus fares, the comfort of the mode of transport, etc.

In this paper, we introduce and study the Traffic Filter Location Problem, where a social planner wants to improve the viability of a transportation network through the introduction of *traffic filters* (sometimes called *bus gates*). In particular, the social planner can sway the traffic flow generated by cars by forbidding the access of some roads to these specific vehicles while leaving them still accessible to public transport, thus encouraging the usage of public transport. Traffic filters simply restrict certain types of road users – usually private vehicles – from driving on certain roads. These policies are not only simple to implement in practice but also effective, as demonstrated by their implementation in numerous British cities, such as Oxford [29]. We show that our approach gives rise to a novel solution concept that can still be interpreted as a Nash Equilibrium of a congestion game and that is consistent with the Discrete Choice Theory framework.

1.1 Our Contribution

In this paper, we formally introduce the Traffic Filter Location Problem (TFLP); design a novel non-atomic multi-modal stochastic equilibrium model with elastic demands; prove that an equilibrium always exists in our model under minimal assumptions (our main theoretical result); demonstrate our formulation can be used to study the TFLP in practice; and run extensive experiments, discussing how the results may inform social planners. This paper provides a framework for social planners and researchers in the field of transportation planning to formally study the TFLP, as well as providing a starting point in terms of a nuanced traffic equilibrium model. We initiate the discussion through the analysis of our experimental results, and what they mean for social planners intending on implementing traffic filters.

In Section 2, we introduce the Traffic Filter Location Problem, formalising it under minimal behavioural assumptions. The core challenge lies in accurately

capturing both the micro-level decisions of individual agents and the macro-level ripple effects on the wider network, caused by the introduction of the filter.

In Section 3, we introduce our traffic model. To properly evaluate traffic filter placements, we require a sophisticated model of agent preferences, decision making, and route selection behaviour, on both individual and system levels. We integrate elastic demands with Random Utility Theory (RUT), allowing agents to choose (i) whether or not to travel, (ii) which mode of transport to travel by, and (iii) the exact route they travel by to reach their destination. We also describe the conditions that would see the emergence of an equilibrium.

In Section 4 we prove the existence of an equilibrium under our model, assuming only continuity of the relevant functions. Informed by the methodology of Braess and Koch [6], we relate the existence of an equilibrium for our model to a fixed point problem.

In Section 5, we present a comprehensive set of experiments to evaluate the performance of our model across a range of parameter configurations and objective functions. We compute the equilibrium in practice by building on a gap function-based column generation algorithm by Lu et al [24]. Our analysis reveals that even on small networks, our formulation offers valuable insights into the behaviour of agents after the introduction of traffic filters. Some of the observed behaviours demonstrate the reverse case of Braess’ Paradox [7], where the addition of a restriction unexpectedly improves overall network performance. Notably, we also find that the stronger agents’ preferences for cars over buses, the more effective the traffic filter measures are. These findings underscore the subtle and often counter-intuitive effects that traffic filters can have on agent dynamics and network efficiency.

Finally, Section 6 provides concluding remarks and suggestions for future works.

1.2 Related works

Optimizing traffic flow on a network stands as a timeless and fundamental challenge. Given its pervasive impact on urban mobility and economic efficiency, researchers and practitioners have long sought effective solutions to reduce congestion and enhance transportation systems’ performance. The task of computing the distributions of flows through a network constitutes the Traffic Assignment Problem (TAP), which traces its origins back to Wardrop’s seminal 1952 paper [44], where the concepts of the User Equilibrium (UE) and System Optimal (SO) are established. One of the most common approaches to the TAP is through considering the emergence of an equilibrium, created by self interested agents, a well studied sub-problem referred to as the Traffic Equilibrium Problem (TEP) [37].

Much work on the TEP is focussed on improving the equilibrium models which determine how traffic is assigned [23,21]. Naturally, many variations of the problem have been explored. For example, some consider agents’ abilities to choose between modes of transport [12,1,14] (the ‘Multi-modal’ variant), and others introduce stochasticity into the agents’ utility functions [31,4], leading to Stochastic User Equilibria (SUE). The study of agents’ decision-making in

the context of the TEP often draws on Discrete Choice Theory and Random Utility Theory, which originate from economics and behavioural science, and model individual decision-making when choosing from a finite set of alternatives [40,5,25].

One of the most important results in the study of the TEP and transportation systems in general is the ‘Price of Anarchy’ [34,33], measuring the inefficiency of a selfish User Equilibrium compared to the ideal System optimal. As a result, a significant body of research examines how the behaviours of agents can be influenced to improve the quality of the resultant equilibrium. This can be achieved by mitigating how intersecting routes affect the travel time of commuters [39], by introducing different protocols to handle such intersections [32,13], or managing traffic lights efficiently [2,28]. Other studies focus on more direct control over agents, controlling autonomous vehicles through motion planning techniques [16]. Indeed, being able to control some of the vehicles traversing the network leads to significant improvements [38,45].

Different work-lines proposed tolling methods to diminish this negative effect. As an example, we have the marginal cost tolls (MCT) in which each agent is required to pay a fee depending on the route they follow [30]. The MCT, however, are hard to compute. For this reason, alternative tolling methods have been proposed as micro-tolling [27,35] and Δ -tolling [36], or more recently the task has been approached through reinforcement learning [10]. Many studies have confirmed that adding tolls have a positive impact on the network [15], however, adding such a tolling system to a city is impractical. Indeed, cities often aim to mitigate congestion by implementing various measures such as providing alternative transportation options like bus lines. This has lead to research into the optimal location of bus dedicated bus lanes [3,42]. Some cities have gone a step further, and begun introducing traffic filters, preventing certain modes of transport from using particular roads [29]. When modelling transportation networks as graphs, traffic filters can be modelled as removing edges, thus linking to the study of edge deletion and network resilience problems [8,9,17], and the study of edge connectivity [26,19,18], both of which have a rich basis in graph theory.

2 The Traffic Filter Location Problem

In this section, we describe what we refer to as the Traffic Filter Location Problem (TFLP) in its general form, making minimal assumptions. In the Traffic Filter Location Problem, a social planner wants to find the best location in a transportation network to locate a traffic filter. The traffic filter is placed on a street, and prevents certain modes of transport from using it. When modelling the network of streets as a graph, this equates to precluding some modes of transport from using a particular edge. The goal is to encourage some modes of transportation by making them more appealing – i.e. increasing the relative utilities of some modes over others. However, the complexity of this problem arises in accurately predicting the reactions of agents in the network. As the

purpose is to encourage a change of strategy, the strategic behaviour of agents must be modelled comprehensively. Table 1 lists some of the variables commonly referenced in this paper. Due to the complexity of the problem, there are several nested definitions. These are laid out together in Section 3.3.

2.1 Transport Network Formulation

We consider a set of points of interest V , that the agents wish to travel between. The agents may travel between these points by a mode of transport m , from the set of available transport modes M . Each mode of transport has an associated set of streets E_m which connect the points of interest. Thus, we can form a directed graph $G_m(V, E_m)$ by considering the points of interest V as vertices and the streets E_m as edges. We assume each of these graphs to be connected, so there always exists a path between every pair of vertices. Multiple modes of transport may use the same edges, that is $|E_{m_1} \cap E_{m_2}| \geq 0$. For example: in a transport network consisting of cars and trams, some edges may represent streets with both tram tracks and lanes for cars. The graphs for each mode of transport are collected into the set $\mathcal{G} := \{G_m | m \in M\}$.

For every source-destination pair of points of interest $(x, y) \in V^2$, there is some number of agents who have a desire to travel between them. This is the underlying or *latent* demand, given by the matrix R , where $R_{x,y} \in \mathbb{R}$. Note that we model agents non-atomically – meaning we consider agents in aggregate, and not as discrete entities. We assume the $R_{x,y}$ is known for all pairs $x, y \in V$ and fixed. When travelling between x and y using a mode of transport m , there may be several different paths available to the agents. For example, an agent waiting at a bus stop may have the choice of taking one of multiple different bus lines to reach their destination. We let $\mathcal{P}^{(m)}$ be the set of all paths across the graph G_m , and $\mathcal{P}_{x,y}^{(m)}$ be the set of all paths starting at vertex x and ending at vertex y . As agents travel along the edges that make up a path, they contribute to the volume of flow on each of those edges. The higher the flow on an edge, the more congested it becomes, slowing down the movement of traffic. This effect is called *latency* and described through a latency function – sometimes called a ‘time delay function’ or ‘volume-delay function’ [20]. The latency on edge e is

Table 1: Commonly Used Variables

Symbol	Description
$G(V, E)$	Graph G with vertices V and edges E
M	Set of modes of transport
m	A given mode of transport
(x, y)	Source-destination pair of vertices
$R, R_{x,y}$	Matrix of latent demands
$D, D_{x,y}^{(m)}$	Matrix of induced demands
U_m	Agent’s utility function for mode m
l_e	Latency function for edge e
f_e	Flow on edge e
$T, T_{x,y}^{(m)}$	Matrix of total latencies
Ω	Objective Function
Γ	Equilibrium Function
$\mathcal{N}$	Network Instance
$\mathcal{P}, \mathcal{P}_{x,y}^{(m)}$	Set of all possible paths on a graph
P	A given path
λ, λ_P	Decision vector / variables

then given by the latency function $l_e(f_e) \rightarrow \mathbb{R}$, where $f_e \in \mathbb{R}$ is the volume of traffic flow over the edge. We assume the l_e to be continuous and monotonically increasing. All of the latency functions for a mode m are given by the vector $L_m := \{l_e\}_{e \in E_m}$, and are each collected into the set $\mathcal{L} := \{L_m | m \in M\}$.

Lastly, each agent travelling following a path P accrues a utility, which we denote as $U_m(t)$, that depends on (i) the mode of travelling m ; (ii) the total latency t of the path taken, which is computed as the sum of the latencies of each edge in the path, so that $t = \sum_{e \in P} l_e(f_e)$; and (iii) a random variable, namely ζ , that models unforeseeable factors. The main assumptions we make on $U_m(t)$ is that its expected value with respect to ζ is finite for every mode of transportation m , i.e. $\mathbb{E}_\zeta[U_m(t)] < +\infty$, and that the function $t \rightarrow \mathbb{E}_\zeta[U_m(t)]$ is a continuous and monotonically decreasing function. In particular, we have that, for a given mode of transport, a path with a lower latency will always have an equal or greater expected utility than a path with a higher latency.

We can now formally define a transportation network instance.

Definition 1 (Network Instance). *A multi-modal transportation network is characterised by the tuple $\mathcal{N} := (M, \mathcal{G}, R, \mathcal{U}, \mathcal{L})$, consisting of a set of modes of transport M , a set of graphs for each mode of transport $\mathcal{G}$, a latent demand matrix R , a set of utility functions $\mathcal{U}$, and a set of latency functions $\mathcal{L}$.*

2.2 User Equilibrium

Agents are selfish, and so will select the combination of transport mode m^* and path P^* which yields the maximal expected utility. Thus, for an agent travelling between x and y ,

$$m^*, P^* = \arg \max_{m \in M, P \in \mathcal{P}_{x,y}^{(m)}} \mathbb{E} \left[U_m \left(\sum_{e \in P} l_e(f_e) \right) \right]. \quad (1)$$

This condition suggests the formation of a *Nash Equilibrium*, when no agent can change their strategy in a way that improves their utility. In transportation modelling, this is specifically a *User Equilibrium* [44] and the task of finding such an equilibrium is the Traffic Equilibrium Problem (TEP) [22]. The complexity of the problem stems from the fact that when an agent switches mode or path, the flows on edges change, which in turn affect the latencies, which then alter the utilities of every agent, meaning more agents may wish to switch strategy. We explore this problem in the following section. However, for now we denote with Γ a function that maps every network instance $\mathcal{N}$ to an equilibrium $\Gamma(\mathcal{N})$ which upholds Condition (1).

2.3 Problem Statement

The purpose of the TFLP is to find the best location for a traffic filter, which prevents some modes of transport for travelling along a street. In this formulation, we say a traffic filter is placed on an edge e and removes it from the graphs of certain modes of transport M' , creating a restricted graph $G'_m(V, E_m \setminus e)$. An edge e is only a valid location for the filter if the graph G'_m remains connected.

Problem 1 (The Traffic Filter Location Problem). Let $(M, \mathcal{G}, R, \mathcal{U}, \mathcal{L})$ be a network instance, and let Ω be an objective function. We assume that Ω depends on the instance $\mathcal{N}$'s corresponding equilibrium state $\Gamma(\mathcal{N})$, taking the form $\Omega(\Gamma(\mathcal{N})) \rightarrow \mathbb{R}$. Then, the TFLP consists of finding the edge e^* to locate a traffic filter such that

$$e^* = \arg \max_{e \in E} \Omega(\Gamma(\mathcal{N}')), \quad \text{where} \quad \begin{aligned} \mathcal{N}' &:= (M, \mathcal{G}', R, \mathcal{U}, \mathcal{L}), \\ \mathcal{G}' &:= \{G'_m(V, E_m \setminus e) \mid \forall m \in M'\}. \end{aligned}$$

Finding the equilibrium $\Gamma(\mathcal{N})$ comprises Problem 2.

3 A Multi-modal Stochastic Equilibrium Model with Elastic Demands

Evaluating the TFLP requires finding an equilibrium $\Gamma(\mathcal{N})$, as stated in Problem 1. However, in order to search for an equilibrium we must first formulate a model for studying multi-modal transport networks. As outlined in Section 1.2, there are many ways to formulate an equilibrium model. We take well establish components from existing formulations and combine them in to produce a novel formulation. Some of the key components our model uses are as follows

Stochastic Utilities. We associate a utility function with each mode of transport, used by agents to select which to travel by. Our model expects an aleatory component to these utilities, although we only specify the aleatory component must be distributed continuously – unlike many existing approaches in the literature, we do not assume a particular logit- or probit-based approach. The use of stochastic utilities will lead to a particular type of equilibrium, known as a Stochastic User Equilibrium (SUE). Additionally, stochastic utilities allows us to model the percentage of agents making a particular choice as the probability of any one agent choosing that strategy.

Elastic Demand. In order to properly capture all the full decision space available to agents, agents should be able to choose not to travel. We implement by effectively treating ‘not travelling’ an special mode of transport and associating a utility with it.

Modes of Transport. For the sake of simplicity, our model assumes there are two transportation modes: public and private. We refer to private means of transportation as *cars*, and public means of transportation as *buses*. When specified, we include ‘not travelling’ as a special mode of transport, denoted with a $\emptyset$.

Path-based Routing When travelling between points of interest, agents consider all available paths between them and choose the one which maximises their utility – i.e. the path with the lowest latency.

We assume that public modes of transport have fixed timetables, and thus contribute a fixed volume of flow, regardless of ridership. We also assume that the latency functions associated with each edge l_e account for this fixed flow, so we

define the flow f_e as a function only of the amount of car traffic using the edge. In effect, $e \notin E_{car} \implies f_e = 0$. We require that journeys are carried out by a single mode of transportation for start to finish. We also assume that the buses have unlimited capacity, and we do not bus lines as distinct from one-another, instead assuming agents can travel by any path through the graph G_{bus} .

Note that most of the important definitions introduced in this section are displayed together in 3.3.

3.1 Stochastic Utilities and Elastic Demand

To design the agent's utility function, we use a Random Utility Model. This is expressed as the sum of a deterministic term and an aleatory term ζ . Our model does not assume a specific distribution for ζ , albeit we do require ζ to be an absolutely continuous probability distribution.

We model the choice of not travelling as a special mode of transport, denoted '0'. As we have reduced the modes of transport to cars and buses, we let $M = \{car, bus\}$. The utility functions for the different modes of transport are defined as follows

$$\begin{aligned} U_m(t) &:= \zeta_m - \alpha_m - \beta_m t \quad \text{for all } m \in M, \\ U_0(t) &:= \zeta_0. \end{aligned} \tag{2}$$

where, for mode of transport m : ζ_m and ζ_0 are random variables representing unknown factors; α_m is a fixed cost; and β_m is a cost per unit latency, effectively a time-cost. By condition (1), we know that agents will always choose to travel by the path that maximises their utilities. It follows that for a fixed mode of transport m , agents will only travel by the available path(s) that have the minimal latency. Therefore the latency $T_{x,y}^{(m)}$ for agents travelling between vertices x and y by mode m can be given

$$T_{x,y}^{(m)} := \min_{P \in \mathcal{P}_{x,y}^{(m)}} \sum_{e \in P} l_e(f_e). \tag{3}$$

By convention, we set $T_{x,y}^{(0)} := 0$. From the perspective of a single agent, the choice of mode of transport for the route x to y is simply given by

$$\arg \max_{m \in M_0} \mathbb{E}[U_m(T_{x,y}^{(m)})].$$

where we let $T^{(m)}$ be a matrix $[T_{x,y}^{(m)}]_{(x,y) \in V^2}$, and we let M_0 be the set of modes of transport M including not travelling '0', $M_0 := M \cup \{0\}$.

We model the elasticity of demand by considering two types of demand: *latent* and *induced*. Recall, the latent demand $R_{x,y}$ is the number of agents that wish to travel between x and y . We define the induced demand $D_{x,y}$ to be the number of agents that actually choose to travel between x and y , where by definition $0 \leq D_{x,y} \leq R_{x,y}$. We let the induced demand for mode m be given by $D_{x,y}^{(m)}$.

We describe the relationship between the latent and induced demands as the probability that the utility of travelling by any mode is higher than the utility of not travelling

$$D_{x,y} := R_{x,y} \Pr \left[\max_{m \in M} U_m(T_{x,y}^{(m)}) > U_0(T_{x,y}^{(0)}) \right], \quad (4)$$

$$D_{x,y}^{(m)} := R_{x,y} \Pr \left[U_m(T_{x,y}^{(m)}) \geq \max_{m' \in M_0 \setminus m} U_{m'}(T_{x,y}^{(m')}) \right]. \quad (5)$$

We let D be the matrix of induced demands $D := [D_{x,y}]_{(x,y) \in V^2}$, and similarly $D^{(m)}$ be the matrix of induced demands for mode m , $D^{(m)} := [D_{x,y}^{(m)}]_{(x,y) \in V^2}$.

3.2 Congestion on the Graph

Owing to our assumptions, the agents travelling between the same source and destination may travel by different paths. We let λ be a vector of decision variables $\{\lambda_P\}_{P \in \mathcal{P}}$, where λ_P represents the fraction of agents travelling between x and y by car that choose to do so via path $P \in \mathcal{P}_{x,y}^{(car)}$,

$$\lambda_P := \mathbb{E}[\text{fraction of } D_{x,y}^{(car)} \text{ using path } P]. \quad (6)$$

Recall that only cars contribute flow to edges, as the flow contributed by public transport is assumed to be fixed and accounted for by the latency functions. Additionally, since we assumed that public modes have unlimited capacity, there is no need for us to consider the exact path agents take via public modes. With λ , we can now specify the flow over an edge e as

$$f_e := \sum_{(x,y) \in V^2} \sum_{P \in \mathcal{P}_{x,y}^{(car)} | e \in P} \lambda_P \cdot D_{x,y}^{(car)}. \quad (7)$$

3.3 Model Formulation and Feasibility

We can now lay out the model formulation in full, and define what constitutes a feasible solution vector λ .

Definition 2. A vector λ is feasible if and only if

$$\forall (x,y) \in V^2 \quad \sum_{P \in \mathcal{P}_{x,y}^{(car)}} \lambda_P = \frac{D_{x,y}^{(car)}}{R_{x,y}}, \quad (8)$$

where for all $(x,y) \in V^2$, $m \in M$, $e \in E$,

$$D_{x,y}^{(m)} := R_{x,y} \Pr \left[U_m(T_{x,y}^{(m)}) \geq \max_{m' \in M_0 \setminus m} U_{m'}(T_{x,y}^{(m')}) \right],$$

$$U_m(t) := \zeta_m - \alpha_m - \beta_m t,$$

$$U_0(t) := \zeta_0,$$

$$T_{x,y}^{(m)} := \min_{P \in \mathcal{P}_{x,y}^{(m)}} \sum_{e \in P} l_e(f_e) := \min_{P \in \mathcal{P}_{x,y}^{(m)}} L_P.$$

$$f_e := \sum_{(x,y) \in V^2} \sum_{P \in \mathcal{P}_{x,y}^{(car)} | e \in P} \lambda_P \cdot D_{x,y}^{(car)},$$

Constraint (8) is obtained by considering (6), which defines λ_P as the expected fraction of agents travelling by car from x to y that use path P . Thus summing λ_P over all paths between x and y will yield the expected fraction of agents travelling from x to y that are using cars, which can be given as $\frac{D_{x,y}^{(car)}}{R_{x,y}}$. Note that $\frac{D_{x,y}^{(car)}}{R_{x,y}}$ is equivalent to the probability of an agent choosing to travel by car for the journey from x to y , which can be obtained by rearranging Definition (5) as $\frac{D_{x,y}^{(car)}}{R_{x,y}} = \Pr[\dots]$.

3.4 Properties of the Equilibrium

Our definitions thus far have created a series of dependencies between different variables. In this subsection, we will demonstrate that these dependencies effectively form a cycle, allowing us to consider the induced demand matrix D as a function of the total latency matrices $T := (T^{(car)}, T^{(bus)})$, and vice versa.

Specifically, for all $m \in M$, Equation (4) shows that the latency matrices $T^{(m)}$ serve as inputs to the utility functions U_m , which themselves determine the induced demand matrix D . Providing us the mappings $T \rightarrow U \rightarrow D$. Similarly, given a demand matrix D , Equation (7) allows us to determine the flows on each edge f_e . Knowing the flows on each edge e , we determine the edge-latencies l_e , which allow us to generate the latency matrices $T^{(m)}$ via Equation (3). We now have a cyclic sequence of mappings $D \rightarrow f_e \rightarrow l_e \rightarrow T \rightarrow U \rightarrow D$, which means that any of the variables involved is a function of any other. We choose to consider the induced demands D and the total latencies T . We let ϕ be a function $\phi : D \rightarrow T$ and ψ be a function $\psi : T \rightarrow D$. The relationships between variables we have described are shown in Figure 1.

If $\psi(D) = T$ and $\phi(T) = D'$ where $D \neq D'$, the solution is unstable. A stable solution, that is an *equilibrium*, will arise if and only if the T induced by D itself induces the same original T . We can now define an equilibrium under our formalism.

Definition 3. *The demand matrix D and the travel time matrices $T := (T^{(car)}, T^{(bus)})$ form an equilibrium under our model if the following condition holds*

$$D = \psi(T) \quad \text{and} \quad T = \phi(D). \quad (9)$$

Recall Problem 1, where we defined Γ to be a function that computes the equilibrium associated with a network instance. We can now formalise this into the following problem.

Problem 2. (Equilibrium) Let $\mathcal{N} := (M, \mathcal{G}, R, \mathcal{U}, \mathcal{L})$ be a network instance. The equilibrium problem consists in finding a function $\Gamma(\mathcal{N}) \rightarrow (D, T)$, such that the couple (D, T) satisfies condition 9, and thus forms a User Equilibrium.

4 Equilibrium Existence

In what follows, we show that Problem 2 always admits a solution, i.e. a couple (D, T) for which Condition (9) holds. To do that, we will show that both ϕ

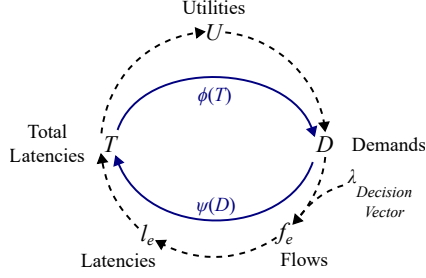


Fig. 1: The dependencies between the main variables involved in computing and equilibrium. Dotted arrows show relationships stated in our formulations, and solid arrows show the functions ϕ and ψ .

and ψ are continuous functions, thus the function $\phi \circ \psi$ is continuous as well. To conclude the proof, we show that $\phi \circ \psi$ admits a fixed point. For the sake of simplicity, we expose our argument for the complete graph $G = (V, E)$, but the same argument can be applied to any connected subgraph of G . This proof follows the general approach first proposed by Braess and Koch [6].

The proofs of Lemmas 1 and 2 are provided in Appendix A.

4.1 From Demands to Total Latencies

Notice that the set of feasible D is compact, as every entry of the matrix is positive and bounded from above by a positive constant (i.e. the total amount of agents). We denote with $\phi : D \rightarrow T$ the function that maps any demand matrix D to its associated time cost matrices T . In a transportation network, if a single agent chooses to change modes of transport, the effect on the total congestion in the network will be very small. However, as more agents switch modes, the effect on congestion will become more pronounced. Therefore, we show that ϕ is continuous as a first step toward our existence result.

Lemma 1 ($\phi(D)$ is continuous). *Given a Congestion Game described as in Section 3.2, let ϕ be the map that assigns the induced demand matrix D to the latency matrices T associated with the Equilibrium of the game induced by the demand D . We then have that if the latency functions l are continuous, the map ϕ is continuous.*

4.2 From Total Latencies to Demands

Let us now consider the reverse case, where we are given the latency matrices T . We denote by $\psi : T \rightarrow D$ the function that maps every latency matrix T to the induced demand D , which is implicit in Definition (4). It is obvious that when planning a journey a small shift to the relative travel times along different routes will cause a small shift in the number of agents taking each route, and that no one small change to travel times would cause a sudden, large change in which routes agents take. As per ϕ , we show that ψ is continuous under minimal assumptions.

Lemma 2 ($\psi(T)$ is continuous). *The function $\psi(T) = D$ that maps every latency matrix to its induced demand is continuous as long as the random variable ζ in (2) is absolutely continuous.*

4.3 The Existence Result

Lastly, we show that a solution always exists via a fixed-point argument. Indeed, any valid solution to our problem is by definition a fixed point of the function $\psi \circ \phi$. The proof hinges upon the continuity of the functions ψ and ϕ that map a latency matrix T to a demand matrix D and vice-versa.

Theorem 1. *Given a connected graph G , there always exist a couple (D, T) that satisfies Condition (9) as long as ϕ and ψ are continuous. In particular, the TFLP always admits a solution.*

Proof. First, we denote with $\mathcal{D}$ the set of all the possible demand matrixes. Since $\mathcal{D}$ is closed and bounded, we have that $\mathcal{D}$ is compact. Let us consider the function $Q : \mathcal{D} \rightarrow \mathcal{D}$ defined as the composition of ψ with ϕ , that is $Q = \psi \circ \phi$. By hypothesis, we have that Q is continuous. Moreover, Q maps a compact set into itself. By the fixed point theorem, we then have that there exists a demand matrix D such that $Q(D) = D$. Let us then set $T = \phi(D)$. By definition of Q , we have that $\psi(T) = D$, which concludes the proof.

5 Experiments

In this section, we describe how we set up our experiments on the TFLP, and report our results. For these experiments, we solve Problem 1 with a brute force search, placing the traffic filter from every edge in turn. We sketch our solving routine in Algorithm 1.

Algorithm 1 Evaluation Procedure for TFLP

- 1: Given a combination of parameters, generate a TFLP instance
 - 2: **for all** valid edges e in the car graph **do**
 - 3: Remove edge e from the car graph
 - 4: Compute the User Equilibrium
 - 5: Evaluate each of the objective functions on the equilibrium
 - 6: **end for**
-

5.1 Parameters

In Definition 3, we define an instance of the TFLP as being given by a network instance $\mathcal{N} := (M, \mathcal{G}, R, \mathcal{U}, \mathcal{L})$ and an optimisation function Ω . The parameters to consider are therefore: (i) the initial graphs $\mathcal{G}$; the latent demand matrix R ; (ii) the latency functions $\mathcal{L}$; (iii) the values which parametrise the utility functions $\mathcal{U}$, namely α_m and β_m , for $m \in \{bus, car\}$, and ζ_m for $m \in \{bus, car, 0\}$; and (iv) the objective function Ω .

To ensure the robustness of our experiments, we vary each of these parameters, and run our experiments on a range of networks. Our tests include a large real-world network from the city of Anaheim³, pictured in Figure 2, as well as two small synthetic networks, named *Quad Grid* and *Basic City*. The section of Anaheim included in the dataset enjoys full coverage by bus routes in the real world, and we likewise assume that the bus- and car- graphs are initially the same in our synthetic tests, i.e. $G_{car} = G_{bus}$. The Anaheim network includes numbers of trips made between parts of the network, allowing us to initialise R directly. For our latency function, we use the widely adopted BPR function $T = T_0(1 + \alpha(\frac{V}{C})^\beta)$ [20]. The Anaheim dataset provides the parameters for this function, fitted based on observed traffic behaviours. In our synthetic tests, we set $\alpha = 1$, $\beta = 2$, and assume for every edge e , $C = 1$ and T_0 is the length of the edge $|e|$, yielding the function $l_e(f_e) := |e| + f_e^2$. We use five sets of parameters for the utility functions, representing different behaviours of agents, ranging from ‘strongly prefer buses’ to ‘strongly prefer cars’. We model the ζ ’s as normal distributions, which allows us to use their CDF functions to directly compute the probabilities given in Equation (4).

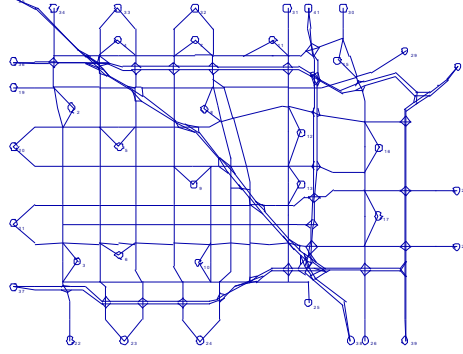


Fig. 2: The Anaheim network, showing edges as straight lines and centroids as circles. The dataset includes the number of trips made between pairs of centroids in a given timeframe. Includes 38 centroids, 416 nodes, and 914 edges.

5.2 Evaluation Metrics

We chose to consider three objective functions, each representing a different metric that a social planner may consider important. The definition of the TFLP (Problem 1) only admits a single objective function Ω , so in our experiments we either (i) consider each metric individually, or (ii) perform mixed optimisation, weighting all three functions equally. In our experiments, we analyse how optimising for a single objective affects the other two. Our three objectives are as follows.

Mean Congestion ($\Omega_{congest}$). The average increase in latency for travellers in the network, relative to the latency if there was no traffic S , defined as $\Omega_{congest} :=$

$$\frac{\sum_{(x,y) \in V^2} D_{x,y}^{(car)}(T_{x,y}^{(car)} - S_{x,y}^{(car)}) + D_{x,y}^{(bus)}(T_{x,y}^{(bus)} - S_{x,y}^{(bus)})}{\sum_{(x,y) \in V^2} D_{x,y}^{(car)} S_{x,y}^{(car)} + D_{x,y}^{(bus)} S_{x,y}^{(bus)}},$$

³ The Anaheim dataset and Figure 2 are provided by [41].

where $S_{x,y}^{(m)}$ is the latency of travelling between x and y by mode m when the flow on every edge is 0. By definition $T_{x,y}^{(m)} \geq S_{x,y}^{(m)}$. However, by Problem 1, the definition of the TFLP, the objective function is to be maximised, so we simply maximise the negative congestion $\Omega_{congest}^- := -\Omega_{congest}$.

Percentage Bus Usage (Ω_{bus}). The fraction of agents that choose to travel via bus. By the nature of the TFLP, we expect planning authorities want to increase bus usage.

$$\Omega_{bus} := \frac{\sum_{x,y \in V^2} D_{x,y}^{(bus)}}{\sum_{x,y \in V^2} D_{x,y}}. \quad \Omega_{travel} := \frac{\sum_{x,y \in V^2} D_{x,y}}{\sum_{x,y \in V^2} R_{x,y}}.$$

Percentage of Travellers (Ω_{travel}). The fraction of agents that choose to travel. We assume the social planner does not want to cause too much disruption to journeys, and so wants to keep this value high.

5.3 Computing the Equilibrium

In order to compute the equilibrium, we adapt the gap function-based column generation approach presented by Lu et al [24]. In brief, the algorithm works by considering a substantially reduced subset of all possible paths, initially only one path between each source-destination pair. An outer-loop (column generation) adds promising new paths to the set of active paths, and the inner loop (path swapping) iteratively distributes the agents between the active paths. The key to the algorithm is the *gap function*, which calculates the difference in cost between a given path q and the path with the minimum cost, weighted by the number of agents using path q . The gap function is effectively an error measurement, since at equilibrium all agents should be travelling along the minimum cost paths. The gap function is used to measure convergence, and to inform the path-swapping. Adapting the equation introduced in the paper to our formulation, the gap function is given as

$$\text{Gap}(\lambda^{(i)}) = \sum_{x,y \in V^2} \sum_{q \in \mathcal{Q}_{x,y}} \lambda_q^{(i)} \cdot R_{x,y} \cdot (L_q - \min_{P \in \mathcal{P}_{x,y}} L_P),$$

where $L_P = \sum_{e \in P} l_e(f_e)$ and $\mathcal{Q}_{x,y} \subseteq \mathcal{P}_{x,y}$ is the set of active paths.

However, the original paper has neither multiple modes or elastic demands. So, we introduce a secondary gap function which measures the difference between the current modal assignments and those prescribed by Equation 5

$$\text{MGap}(\lambda^{(i)}) = \sum_{x,y \in V^2} \left(R_{x,y} \cdot \sum_{q \in \mathcal{Q}_{x,y}} \lambda_q^{(i)} \right) - D_{x,y}^{(car)}.$$

The total gap at iteration i is thus simply $\text{Gap}(\lambda^{(i)}) + \text{MGap}(\lambda^{(i)})$. We perform a mode-swapping step in the inner loop immediately before the path-swapping step, following the same scheme.

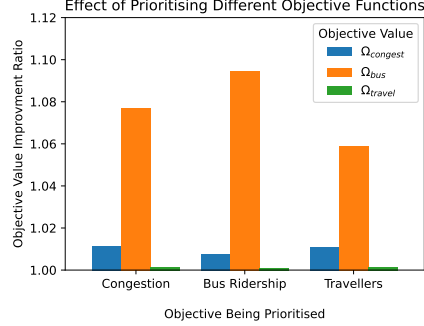


Fig. 3: A comparison of how prioritising each objective function when selecting the edge to remove affects the other objectives. The y-axis shows the improvement in the objective value with respect to the values observed prior to edge removal. (*Anaheim network*)

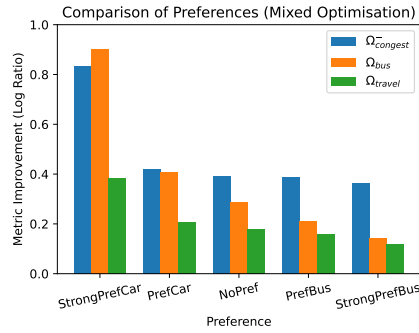


Fig. 4: The improvement in the different objective values after removing an edge with different agent preferences. We observe a greater improvement in all three objectives when the agents prefer travelling by car over bus. (*Quad-grid network*)

5.4 Results

The key findings of our analysis are: (i) Removing a single edge from the network almost always leads to improvements across all three objectives. (ii) The stronger the agents’ preferences for cars, the more the objective values improve after edge-removal. (iii) The three objective functions we consider, $\Omega_{congest}^-$, Ω_{bus} , and Ω_{travel} , are correlated; optimising one will improve the other two (Figure 3).

While it is intuitive that removing edges encourages a shift towards buses, it is less obvious that this benefit outweighs the congestion from diverted traffic. This counter-intuitive result is effectively the inverse of Braess’ Paradox [7]. Rather than added capacity increasing congestion, the removal car capacity leads to reduced congestion. As depicted in Figure 3, in the Anaheim network we see up to a 10% increase in bus usage network-wide from the remove of a single edge. Furthermore, we see a reduced in total congestion of around 1%. The congestion reduction is not caused by agents choosing not to travel, as we observe a slight increase in the fraction of agents choosing to travel.

When agents have a stronger preference for car travel, one might expect edge removals to have less impact, as agents are less willing to switch modes. However, we find the opposite: higher car preference amplifies the benefits of edge removal (Figure 4). This appears to result from increased baseline congestion—greater car usage leads to higher traffic, making bus travel on the removed edge more attractive. Since flow grows linearly with car preference but latency increases

polynomially with flow, congestion effects dominate, strengthening the impact of mode shifts. These findings may inform policy decisions in car-dependent transport networks.

6 Conclusion and Future Work

In this paper, we introduced and formally stated the TFLP. To address it, we proposed a novel multi-modal framework for modelling traffic flow that integrates stochastic equilibria with elastic demand. We proved the existence of an equilibrium under minimal assumptions and demonstrated how it can be computed in practice by adapting a gap function-based column generation algorithm to our framework. This allowed us to analyse the effects of traffic filters on both real and synthetic networks across a wide range of parameters. Our results show that traffic filters consistently reduce congestion, increase bus usage, and boost overall travel demand – effects which are surprisingly increased when agents strongly prefer cars. Overall, our framework offers a robust foundation for evaluating sustainable transport interventions and supporting evidence-based urban mobility policy.

Future work will aim to develop heuristics or algorithms for directly identifying the most effective edge(s) to remove. We also plan to extend the model to account for dynamic traffic patterns and more heterogeneous agent behaviours. In addition, it would be valuable to explore methods for estimating the impact of removing multiple edges simultaneously without requiring full simulation.

Acknowledgments. This work was partially supported by a Leverhulme Trust Research Project Grant (2021 – 2024) and the EPSRC grant (EP/W014912/1).

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A Missing Proofs

In this section, we report the missing proofs.

Proof (Proof of Lemma 1). To prove the theorem it suffices to consider the case in which we have two induced demand matrices, namely D and D' such that they differ only on one entry, namely $(x, y) \in V^2$. Note that the entries of D and D' belong to the set $[0, M]$, where M is a constant that is large enough. We recall that, given the demand matrix D in input, the function ϕ returns $T = (T^{(car)}, T^{(bus)})$, which is the pair of matrices inducing the latency associated with cars and buses. For the sake of simplicity, we limit our discussion to the case in which ϕ returns only the latency of the cars, namely $T^{(car)}$, as the same argument can be applied to the buses latencies.

Let us now take any $\epsilon > 0$, we prove that ψ is continuous by proving that there exists a $\delta \geq 0$ for which it holds

$$|D - D'| < \delta \implies |\phi(D)_{x,y} - \phi(D')_{x,y}| < \epsilon, \quad \forall (x,y) \in V^2,$$

where $|D - D'| = \sqrt{\sum_{x,y} (D_{x,y} - D'_{x,y})^2}$.

By the nature of the Nash Equilibrium formed in the congestion game, the agents travelling between two nodes x and y will only travel on the paths in $\mathcal{P}_{x,y}$ with the minimal latency. We call these paths $\mathcal{P}_{x,y}^{(min)}$. Since all these paths are of equal latency, we can reason about an arbitrary path $P_{x,y} \in \mathcal{P}_{x,y}^{(min)}$. Let f_e and f'_e be the flows on edge e induced by D and D' , respectively, given by Equation 7. $\phi(D)$ and $\phi(D')$ are thus given by

$$\phi(D)_{i,j} = \sum_{e \in P_{i,j}} \int_0^{f_e} l_e(z) dz, \quad \phi(D')_{i,j} = \sum_{e \in P_{i,j}} \int_0^{f'_e} l_e(z) dz.$$

The change in demand is bounded by δ as given by $|D_{x,y} - D'_{x,y}| < \delta$. As the flow is defined as the sum over a multiple of $D_{x,y}$, it follows that the change to the flow over an edge is bounded by $f'_e < f_e + M\delta$ for some constant M . We thus have that the maximum change in latency between any two points in the network i and j can be written

$$\begin{aligned} |\phi(D)_{i,j} - \phi(D')_{i,j}| &= \left| \sum_{e \in P_{i,j}} \int_0^{f_e} l_e(z) dz - \sum_{e \in P_{i,j}} \int_0^{f'_e} l_e(z) dz \right| \\ &= \left| \sum_{e \in P_{i,j}} \left(\int_0^{f_e} l_e(z) dz - \int_0^{f'_e} l_e(z) dz \right) \right|, \end{aligned}$$

since $f'_e > f_e$ and $l_e(z)$ is a non-decreasing function, we have that

$$\begin{aligned}
&= \sum_{e \in P_{i,j}} \left(\int_0^{f'_e} l_e(z) dz - \int_0^{f_e} l_e(z) dz \right) \\
&= \sum_{e \in P_{i,j}} \int_{f_e}^{f'_e} l_e(z) dz \\
&\leq \sum_{e \in P_{i,j}} \int_{f_e}^{f_e + M\delta} l_e(z) dz,
\end{aligned}$$

our model assumes $l_e(z)$ to be continuous, so we can bound the above with a constant multiple of $M\delta$, which we set as $L\delta$, therefore

$$\leq \sum_{e \in P_{i,j}} L\delta,$$

which can clearly also be bounded by constant multiple of $L\delta$, $K\delta$

$$\leq K\delta.$$

Thus, if we take δ to be $\frac{\epsilon}{K}$, we get

$$|\phi(D)_{i,j} - \phi(D)_{i,j}| < \epsilon,$$

which concludes the proof.

Proof (Proof of Lemma 2). By definition, we have that ψ is such that

$$D_{x,y} = \psi(T)_{x,y} = \Pr[U_{car}(T_{x,y}^{(car)}) \geq \max_{m' \in M_0} U_{m'}(T_{x,y}^{(m')})]$$

Without loss of generality, let us fix a couple of nodes x, y . First, we notice that, by assumption, all the random variables U_m are absolutely continuous, thus the random variable $M(T_{x,y}) = \max_{m' \in M_0} U_{m'}(T_{x,y}^{(m')})$ is absolutely continuous as well. Likewise, we have that $\Delta U(T_{x,y}) := \zeta_{car} - \alpha_{car} - \beta_{car} T_{x,y}^{(car)} - M(T_{x,y})$ is an absolutely continuous variable for every $T_{x,y}$.

Notice that, within this notation, we have that $\psi(T_{x,y}) = 1 - F_{\Delta U(T_{x,y})}(0)$, we then need to show that the function

$$T \rightarrow F_{\Delta U(T_{x,y})}(0)$$

is continuous. To do that, we show that given T and $\epsilon > 0$, there exists a $\delta > 0$ for which

$$|T - T'| \leq \delta \implies |F_{\Delta U(T_{x,y})}(0) - F_{\Delta U(T'_{x,y})}(0)| \leq \epsilon.$$

We recall that $T = (T^{(car)}, T^{(bus)})$, hence the condition $|T - T'| \leq \delta$ implies

$$|T^{(car)} - T'^{(car)}| \leq \delta \quad \text{and} \quad |T^{(bus)} - T'^{(bus)}| \leq \delta. \quad (10)$$

Consequently, we have that, if $|T - T'| \leq \delta$,

$$\left| \max\{\zeta_{bus} - \alpha_{bus} - \beta_{bus} T^{(bus)}, \zeta_0\} - \max\{\zeta_{bus} - \alpha_{bus} - \beta_{bus} T'^{(bus)}, \zeta_0\} \right| \leq \delta$$

for every realization of ζ_{bus} and ζ_0 . By definition of $\Delta U(T_{x,y})$, we then have that

$$|\Delta U(T_{x,y}) - \Delta U(T'_{x,y})| \leq K\delta,$$

for every realization of the random variables $\Delta U(T_{x,y})$ and $\Delta U(T'_{x,y})$. In particular, we conclude that, if $|T - T'| \leq \delta$, then

$$\begin{aligned} P(\Delta U(T_{x,y}) > 0) - P(\Delta U(T_{x,y}) > -K\delta) \\ \leq P(\Delta U(T_{x,y}) > 0) - P(\Delta U(T'_{x,y}) > 0) \end{aligned}$$

and

$$\begin{aligned} P(\Delta U(T_{x,y}) > 0) - P(\Delta U(T'_{x,y}) > 0) \\ \leq P(\Delta U(T_{x,y}) > 0) - P(\Delta U(T'_{x,y}) > K\delta) \end{aligned}$$

therefore

$$\begin{aligned} |P(\Delta U(T_{x,y}) > 0) - P(\Delta U(T'_{x,y}) > 0)| \\ \leq \max\{|P(\Delta U(T_{x,y}) > 0) - P(\Delta U(T'_{x,y}) > \pm K\delta)|\}. \end{aligned}$$

Lastly, notice that, since ΔU is absolutely continuous random variable, we have that for a suitable constant $C > 0$ it holds

$$\max\{|P(\Delta U(T_{x,y}) > 0) - P(\Delta U(T_{x,y}) > \pm K\delta)|\} \leq CK\delta.$$

Since

$$\begin{aligned} |F_{\Delta U(T_{x,y})}(0) - F_{\Delta U(T'_{x,y})}(0)| \\ \leq \max\{|P(\Delta U(T_{x,y}) > 0) - P(\Delta U(T_{x,y}) > \pm K\delta)|\}, \end{aligned}$$

we conclude the proof by taking δ such that $\epsilon = CK\delta$.